

Cardiovascular Risk Prediction Using Machine Learning

I. LITERATURE REVIEW

Cardiovascular risk prediction has been widely studied using traditional statistical models, machine learning algorithms, and more recent deep learning and ensemble approaches. Previous studies provide useful guidance for selecting suitable risk factors, prediction models, validation methods, and evaluation measures for cardiovascular risk assessment.

Hughes et al. developed SEER, a deep convolutional neural network that predicts long-term cardiovascular risk from 12-lead ECG signals, achieving an AUC of 0.83 for five-year cardiovascular mortality at Stanford [1]. The study demonstrates the potential of deep learning for cardiovascular risk assessment, although its dependence on ECG signals makes it different from our structured clinical and lifestyle data approach.

Mohan et al. proposed a Hybrid Random Forest with Linear Model (HRFLM) and reported 88.7% accuracy for heart disease prediction using the Cleveland dataset [2]. The study is relevant because it compares machine learning methods using common cardiovascular attributes, but its small dataset and disease-classification objective limit its direct application to general cardiovascular risk prediction.

Chicco and Jurman applied several machine learning models to heart-failure survival prediction and identified serum creatinine and ejection fraction as the strongest predictors [3]. Their work demonstrates the importance of feature selection in cardiovascular prediction, although the study used only 299 heart-failure patients and focused on mortality rather than general CVD risk.

Cho et al. compared logistic regression, TreeBag, Random Forest, AdaBoost, and neural networks in 222,998 Korean adults, with the neural network obtaining the highest C-statistic of 0.751 [4]. This study is relevant because it directly evaluates multiple machine learning approaches for future cardiovascular risk, although its population-specific design may limit generalization to other populations.

Li et al. evaluated 19 statistical and machine learning models using approximately 3.6 million patients and found similar population-level C-statistics of around 0.87 despite substantial differences in individual predictions [5]. Their findings show that cardiovascular risk models should be compared using reliability and calibration as well as overall predictive performance, particularly when long-term outcomes are considered.

Khan et al. developed the American Heart Association PREVENT equations using more than six million US adults and obtained external-validation C-statistics of 0.794 for women

and 0.757 for men [6]. The study is relevant because it uses established clinical factors such as blood pressure, cholesterol, smoking, diabetes, and kidney function, although PREVENT is based on survival modeling rather than conventional machine learning classification.

Hippisley-Cox et al. developed QRISK2 using Cox proportional hazards regression on approximately 2.3 million patients in England and Wales and demonstrated improved discrimination and calibration over the modified Framingham score [7]. Several QRISK2 predictors, including age, smoking, blood pressure, cholesterol, BMI, and diabetes, are relevant to our project, although the authors noted possible advantages from validation in a population similar to the derivation cohort.

Barbieri et al. combined deep learning with survival analysis to predict five-year cardiovascular risk in more than 2.1 million New Zealand adults and reported better discrimination and calibration than conventional Cox models [8]. The study supports the use of nonlinear machine learning methods, but its administrative dataset lacked several direct clinical measurements such as blood pressure and lipid values.

Neumann et al. developed a cardiovascular risk model for healthy older adults using Cox regression and LASSO-based variable selection, obtaining an AUC of 0.681 in development and 0.642 in external validation [9]. The study demonstrates the value of selecting clinically relevant predictors, although its restriction to individuals aged 70 years or older limits application to wider age groups.

Alaa et al. applied AutoPrognosis to 423,604 UK Biobank participants using 473 variables and achieved an AUC-ROC of 0.774 compared with 0.724 for the Framingham score [10]. This study shows that automated machine learning and additional predictors can improve cardiovascular risk estimation, although its large number of variables and complex framework may not be practical for smaller clinical datasets.

Unnikrishnan et al. compared the Framingham equation, logistic regression, and Support Vector Machine using cardiovascular health factors, with SVM showing improved sensitivity and specificity [11]. The study is closely related to our project because it uses factors such as smoking, cholesterol, blood pressure, BMI, diabetes, and hypertension, although validation was limited to a single population.

Islam et al. used BDHS 2022 data from 2,221 Bangladeshi individuals and compared logistic regression, Naive Bayes, ANN, Random Forest, and XGBoost for identifying individuals at high cardiovascular risk [12]. XGBoost achieved the best performance with 68.22% accuracy and an AUC of 0.721, while validation on the Framingham dataset reached

86.25% accuracy; however, the cross-sectional design and limited clinical variables restrict longitudinal risk assessment.

Tasnim et al. proposed a regression-based stacking ensemble combining Ridge Regression, Theil-Sen Regression, and Gradient Boosting using clinical data from 1,529 Bangladeshi patients, reporting approximately 96% predictive performance with SHAP-based interpretation [13]. The study is highly relevant to our project because it applies regression, ensemble learning, and explainability to structured cardiovascular risk data, although data from a single institution limit external generalizability.

Van Calster et al. emphasized that discrimination alone is insufficient for clinical prediction and argued that cardiovascular risk models should also be evaluated for calibration [14]. This is relevant to our project because accurate risk probabilities require agreement between predicted and observed risk, while poor calibration may remain unnoticed when only accuracy or AUC is reported.

Shah et al. developed a hybrid ensemble framework combining Gradient Boosting, CatBoost, LightGBM, SVM, neural networks, and XGBoost with explainable AI, achieving an AUC-ROC of 0.82 [15]. The study supports ensemble learning and SHAP-based interpretation for cardiovascular risk prediction, although combining multiple datasets and complex models increases computational and implementation requirements.

Suri et al. reviewed 265 studies involving multiclass, multi-label, and ensemble machine learning approaches for cardiovascular risk stratification [16]. The review found ensemble-based approaches to be widely used and emphasized feature selection, appropriate ground truth, clinical validation, and risk of bias, providing useful methodological guidance for our model development.

Hossain et al. compared logistic regression, Naive Bayes, Decision Tree, AdaBoost, Random Forest, and Bagging Tree using Bangladeshi clinical data, with Random Forest achieving 98.04% accuracy and an AUC of 0.989 [17]. The study is relevant because it examines cardiovascular predictors in a Bangladeshi population, although the outcome was primarily existing CVD status and the sample contained only 651 individuals, limiting its use as a long-term risk prediction model.

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